

ELEC2134 Lecture Notes

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1 Topic 1: Transform Methods

1.1 Waveforms and Periodic Functions

1.1.1 Waveform Fundamentals

Definition 1.1.

- A waveform describes the shape of a signal as a function of time.
- A signal is a function that varies with time.
- A unit step signal stays at 0 until a certain time, then instantly becomes 1.
- A ramp signal increases linearly with time, starting from 0.

The period (T) is the time taken for a waveform to complete one full cycle and start repeating

$$\text{frequency } f = \frac{1}{T} \text{ Hertz (Hz)}.$$

Hertz is cycles per second. The units of period is time.

Example 1.2. Some common frequencies.

Frequency (f)	Period ($f = \frac{1}{T}$)
1 kHz	$1 \times 10^{-3} s = 1ms$
1 MHz	$1 \times 10^{-6} s = 1\mu s$
1 GHz	$1 \times 10^{-9} s = 1ns$
1 THz	$1 \times 10^{-12} s = 1ps$

The peak amplitude of a wave is the distance between the x axis and the highest point of the wave. The peak-to-peak amplitude is two times this amplitude.

Periodic Waveforms (or Functions) 1.3.

A periodic waveform or function is any waveform or function that satisfies

$$x(t) = x(t + T),$$

where t is the time variable, and T is the period. In general,

$$x(t) = x(t + nT), \forall n \in \mathbb{Z}.$$

1.1.2 Sinusoid waveforms

A sinusoidal waveform is a type of period waveform, and can be described using the following equation

$$x(t) = A \sin(2\pi ft),$$

where $x(t)$ is the instantaneous amplitude, A is the peak amplitude, and f is the frequency in Hz. It can also be described in terms of period with

$$x(t) = A \sin\left(\frac{2\pi}{T}t\right),$$

where T is the period. This can be simplified in terms of angular frequency $\omega = 2\pi f = \frac{2\pi}{T}$ radians/sec as

$$x(t) = A \sin(\omega t).$$

1.1.3 Other Types of Waveforms

Square Waveform

A square waveform is used mostly in digital electronic circuits for clock and timing control signals. It has very steep vertical transitions and flat top and bottoms.

It has pulse width $\tau = \frac{T}{2}$ and frequency $f = \frac{1}{T}$.

Rectangular Waveform

A rectangular waveform is similar to a square wave, however the duration of the high level (pulse width) is shorter than half the period. This is described by the duty cycle, which is the amount of time a signal is ON compared to the total time of one cycle.

Triangular Waveform

A triangular waveform is a non-sinusoidal waveform that linearly oscillates between a positive and negative peak. They are usually comprised of many sine or cosine waves with different frequencies and amplitudes.

Sawtooth Waveform

A sawtooth waveform is a periodic waveform whose shape resembles the teeth of a sawblade. It is similar to a triangular waveform, however the decay is more similar to a square wave's.

1.1.4 Harmonic Components

A harmonic is a multiple of the fundamental angular frequency of a waveform. The fundamental frequency is the frequency as defined above, giving the harmonics to be

$$\omega_0 = 2\pi f_0 = \frac{2\pi}{T} \Rightarrow w_n = n\omega_0, f_n = nf_0,$$

where n is a positive integer.

The average value of a waveform is the mean level of the signal over one complete cycle. This is the overall DC component of the waveform. For a periodic waveform $x(t)$ with period T , this is

$$V_{avg} = \frac{1}{T} \int_0^T x(t) dt.$$

1.2 Fourier Series

1.2.1 Frequency Domain Representation

The frequency domain representation of a waveform shows how a time-domain (i.e. amplitude as a function of time) signal is distributed across different frequencies, effectively decomposing a signal into its component frequencies.

Example 1.1. What does the phase-domain of the sinusoid $v(t)$ below look like?

$$v(t) = \frac{4}{\pi} \sin(\omega_0 t) + \frac{4}{3\pi} \sin(3\omega_0 t) + \frac{4}{5\pi} \sin(5\omega_0 t).$$

Answer: This would be a plot with three peaks; one at ω_0 , one at $3\omega_0$, and one at $5\omega_0$. The first one has a height of $4/\pi$, the second $4/3\pi$, and the last $4/5\pi$.

1.2.2 Fourier Series

This process of breaking down a signal is called Fourier decomposition, resulting in a Fourier series. This is a sum of sinusoidal components with different frequencies and amplitudes. The main application of a Fourier series is to determine the harmonic components that are present in a complex periodic waveform.

Any periodic waveform with a period of T seconds can be represented by an infinite series of harmonically related sinusoids. The representation is known as the trigonometric Fourier series.

$$x(t) = A_0 + \sum_{i=1}^{\infty} A_n \cos(n\omega_0 t) + B_n \sin(n\omega_0 t).$$

The fundamental frequency of $x(t)$ is $\omega_0 = 2\pi f_0 = \frac{2\pi}{T}$ radians/sec. The integer multiples of ω_0 are known as harmonic frequencies of $x(t)$. The Fourier coefficients A_0 , A_n and B_n are calculated directly from the signal $x(t)$ using the following equations

$$\begin{aligned} A_0 &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \\ \omega_0 &= 2\pi f_0 = \frac{2\pi}{T} \text{ radians/sec} \\ A_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt \\ B_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt. \end{aligned}$$

Fourier Series of an Even Periodic Function

A function is defined as even if $x(t) = x(-t)$. For these functions/waveforms, the expressions for the Fourier coefficients reduce to

$$A_0 = \frac{2}{T} \int_0^{T/2} x(t) dt, \quad A_n = \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt, \quad B_n = 0.$$

The reason that the B coefficients disappear is because the periodic waveform is even, so there are no terms from the odd sine function.

Fourier Series of an Odd Periodic Function

A function is defined as odd if $x(t) = -x(-t)$. For these function/waveforms, the expressions reduce to

$$A_0 = A_n = 0 \text{ for } n > 0, \quad B_n = \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$$

1.2.3 Reconstructing a Waveform

To reconstruct (or synthesise) a waveform/function from a set of Fourier coefficients, simply plug them back into the above expression for Fourier series along with their index.

1.3 Alternative Trigonometric Form

In the frequency domain, the amplitude of each frequency component is determined by A_n and B_n , with contributions from both the sine and cosine terms. We would now like to express the signal in an alternative form in which the amplitude of each frequency components is obtained using a single cosine function with a phase angle

$$x(t) = K_0 + \sum_{n=1}^{\infty} K_n \cos(n\omega_0 t + \phi_n).$$

This is the amplitude-phase representation. We are able to find the phase angle ϕ_n with

$$\phi_n = \tan^{-1} \left(-\frac{B_n}{A_n} \right) = -\tan^{-1} \left(\frac{B_n}{A_n} \right).$$

With this phase angle, we are also able to express the Fourier series using a complex number with $K_n \angle \phi_n = A_n - jB_n$.

1.4 Complex Form of the Fourier Series

The cosine-sine and amplitude-phase Fourier series from above are all trigonometric Fourier series. Another alternative Fourier series representations uses exponential functions instead of trigonometric functions. This

form is therefore called the complex form of the (or exponential) Fourier series, and is given by

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt,$$

where $\omega_0 = \frac{2\pi}{T}$. The DC component is then given by C_0 , and C_n can be found with

$$C_n = \frac{1}{2}(A_n - jB_n), \quad |C_{-n}| = |C_n| = \frac{1}{2}\sqrt{A_n^2 + B_n^2}.$$

- For even symmetry: $C_n = \frac{2}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt$
- For odd symmetry: $C_n = -\frac{2j}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt$

1.5 Effect of Shifting

Shifting a periodic function along either the vertical or the horizontal axis will generally affect its Fourier coefficients. These are also affected by the presence of certain types of functional symmetries in $f(t)$, particularly even-, odd-, and half-wave symmetry because some coefficient sets vanish altogether.

Shifting a periodic function $f(t)$ along either the vertical or the horizontal axis affects its amplitude-phase Fourier series. It is apparent that shifting $f(t)$ up or down the vertical axis changes only the average value K_0 , leaving the amplitude K_n and phase angle ϕ_n of each harmonics component unaffected.

1.6 Half-Wave Symmetry

An important type of symmetry is the half-wave symmetry. A periodic function is half-wave symmetric if

$$x(t) = -x\left(t - \frac{T}{2}\right).$$

Shifting a half-wave symmetric wave or function by half a period then inverting it yields the original function. If a function has half-wave symmetry, then both A_n and B_n are 0 for even values of n .

1.6.1 Odd Half-Wave Symmetry

A periodic signal has odd half-wave symmetry if both of the following are true:

$$x(t) = -x\left(t - \frac{T}{2}\right), \quad \text{and} \quad x(t) = -x(-t).$$

For a function with odd half-wave symmetry, the coefficients are given by

$$A_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

$$B_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

For odd half-wave symmetry

- All cosine terms are 0 $A_n = 0$.
- All even harmonics are 0.
- Only odd sine harmonics appear in the Fourier series.

1.6.2 Even Half-Wave Symmetry

A periodic signal has even half-wave symmetry if both of the following are true:

$$x(t) = -x\left(t - \frac{T}{2}\right), \quad \text{and} \quad x(t) = x(-t).$$

For a function with even half-wave symmetry, the coefficients are given by

$$A_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$
$$B_n = \begin{cases} \frac{4}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

For even half-wave symmetry

- All sine terms are 0 $B_n = 0$.
- All sine harmonics are 0.
- Only odd cosine harmonics appear in the Fourier series.

1.7 Examples

1.7.1 Waveforms

Example 1.1. Sum these two sine waves

$$f(t) = 10 \sin(200t)$$

$$g(t) = 8 \sin(200t)$$

Answer: These sine waves have the same period $2\pi f = 200$, $f = \frac{200}{2\pi} \text{ Hz} = 31.83 \text{ Hz}$.

A Filters

A.1 Types of Filters

The following are idealised specifications for the most common filter functions

Ideal Low-Pass Filter

Passes low frequencies, blocks high frequencies.

$$|H(j\omega)| = \begin{cases} 1, & 0 \leq \omega \leq \omega_c \\ 0, & \omega > \omega_c \end{cases}$$

Passband: $0 \leq \omega < \omega_c$, stopband: $\omega_c < \omega \leq \infty$.

Ideal High-Pass Filter

Passes high frequencies, blocks low frequencies.

$$|H(j\omega)| = \begin{cases} 0, & 0 \leq \omega < \omega_c \\ 1, & \omega > \omega_c \end{cases}$$

Passband: $\omega_c < \omega \leq \infty$, stopband: $0 \leq \omega < \omega_c$.

Ideal Band-Pass Filter

Passes only frequencies between ω_L and ω_H .

$$|H(j\omega)| = \begin{cases} 1, & \omega_L \leq \omega \leq \omega_H \\ 0, & \text{otherwise} \end{cases}$$

Passband: $\omega_L \leq \omega \leq \omega_H$, stopbands: $\omega < \omega_L$ and $\omega > \omega_H$.

Ideal Band-Stop (Band-Reject) Filter

Rejects frequencies between ω_L and ω_H .

$$|H(j\omega)| = \begin{cases} 0, & \omega_L \leq \omega_H \\ 1, & \text{otherwise} \end{cases}$$

Passbands: $\omega < \omega_L$ and $\omega > \omega_H$, stopband: $\omega_L < \omega < \omega_H$.

B Useful Integrals

B.1 Integrals

For integer $n \neq 0$

$$\begin{aligned} \int \sin(ax) dx &= -\frac{1}{a} \cos(ax) \\ \int \cos(ax) dx &= \frac{1}{a} \sin(ax) \\ \int_0^T \sin(n\omega_0 t) dt &= 0 \\ \int_0^T \cos(n\omega_0 t) dt &= 0 \end{aligned}$$

B.2 Integration by Parts

B.2.1 Key Formula

Key formula:

$$\int u dv = uv - \int v du.$$

$$\int x e^{ax} dx = \frac{e^{ax}}{a^2} (ax - 1)$$

$$\int x^2 e^{ax} dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

B.2.2 Cosine

Key formula:

$$\int t \cos(n\omega_0 t) dt.$$

Let $u = t \Rightarrow dt$, $dv = \cos(n\omega_0 t) dt \Rightarrow v = \frac{1}{n\omega_0} \sin(n\omega_0 t)$

$$\int t \cos(n\omega_0 t) dt = \frac{1}{n\omega_0} \sin(n\omega_0 t) - \frac{1}{(n\omega_0)^2} \cos(n\omega_0 t) + G$$

B.2.3 Sine

Key formula:

$$\int t \sin(n\omega_0 t) dt.$$

Let $u = t \Rightarrow du = dt$, $dv = \sin(n\omega_0 t) dt \Rightarrow v = -\frac{1}{n\omega_0} \cos(n\omega_0 t)$

$$\int t \sin(n\omega_0 t) dt = -\frac{t}{n\omega_0} \cos(n\omega_0 t) + \frac{1}{(n\omega_0)^2} \sin(n\omega_0 t) + G$$

B.2.4 Definite Integration by Parts

For $0 \leq t \leq T$ and integer n

$$\int_0^T t \sin(n\omega_0 t) dt = \left[\frac{T}{n\omega_0} \right] \quad \text{and} \quad \int_0^T t \cos(n\omega_0 t) dt = \boxed{0}.$$

B.3 Definite Integrals

x^2 and trigonometric function:

$$\int_0^T x^2 \cos(nx) dx = \left[\frac{x^2}{n} \sin(nx) + \frac{2x}{n^2} \cos(nx) - \frac{2}{n^3} \sin(nx) \right]_0^T$$

$$\int_0^T x^2 \sin(nx) dx = \left[-\frac{x^2}{n} \cos(nx) + \frac{2x}{n^2} \sin(nx) + \frac{2}{n^3} \cos(nx) \right]_0^T.$$

x and trigonometric function:

$$\int_0^T x \cos(nx) dx = \left[\frac{x}{n} \sin(nx) + \frac{1}{n^2} \cos(nx) \right]_0^T$$

$$\int_0^T x \sin(nx) dx = \left[-\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \right]_0^T.$$

Miscellaneous:

$$\begin{aligned}\int_0^T \cos(m\omega_0 t) \sin(n\omega_0 t) dt &= 0 \quad \text{for all } m \text{ and } n \\ \int_0^T \sin(m\omega_0 t) \sin(n\omega_0 t) dt &= \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases} \\ \int_0^T \cos(m\omega_0 t) \cos(n\omega_0 t) dt &= \begin{cases} 0 & \text{for all } m \neq n \\ \frac{T}{2} & \text{for } m = n \end{cases}.\end{aligned}$$

C Trigonometric Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta)$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha.$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$= 1 - 2 \sin^2 \alpha$$

$$= \cos^2 \alpha - \sin^2 \alpha$$

$$\sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha)$$

$$\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$$

$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{2j}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left(\alpha + \tan^{-1} \left(\frac{B}{A} \right) \right).$$